

SIMATS ENGINEERING

ASSESSMENT TOOL – 3

Course Name: Internet Programming

Course Code: CSA4305

Course Outcome Covered: CO-1

Assessment Tool: Code Along Exercise – Make Your First HTML Page with CSS

Weightage: 10%

Total Marks: 10

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Scenario: Web Development Standards Implementation

A startup company is developing a modern web application. During development, the team faces issues related to video embedding, secure form submission, semantic HTML, XML syntax, and web communication protocols.

The team must select the correct technologies and implement them according to web development standards.

1. Correct HTML Tag for Video Embedding

Given Options

A.

`<media src="video.mp4"></media>`

B.

`<video src="video.mp4" controls></video>`

C.

`<movie src="video.mp4"></movie>`

D.

`<vid src="video.mp4"></vid>`

Correct Answer

Option B:

`<video src="video.mp4" controls></video>`

Justification

The <video> element is the standard HTML5 element used to embed video content in a web page.

The src attribute specifies the location of the video file, while the controls attribute provides controls such as:

- Play
- Pause
- Volume control
- Progress bar
- Full-screen option

The tags <media>, <movie>, and <vid> are not standard HTML5 elements for displaying videos.

Improved Video Implementation

```
<video width="640" height="360" controls>
```

```
  <source src="video.mp4" type="video/mp4">
```

```
  <source src="video.webm" type="video/webm">
```

```
  Your browser does not support the video element.
```

```
</video>
```

This implementation provides multiple video formats for better browser compatibility.

2. Appropriate HTTP Method for Form Submission

Given Options

- A. GET
- B. POST
- C. PUT
- D. DELETE

Correct Answer

Option B: POST

Explanation

The POST method is suitable for submitting registration form data to the server.

The data is sent inside the HTTP request body instead of being displayed in the browser URL.

Example:

```
POST /register HTTP/1.1
```

```
Content-Type: application/x-www-form-urlencoded
```

```
name=Mathusri&email=mathusri@example.com&password=12345
```

POST is preferred because:

- User data is not shown in the URL.
- It supports larger amounts of data.
- It is suitable for creating new user records.
- It avoids storing sensitive form data in browser history.
- It works securely when used with HTTPS.

GET should not be used for sensitive registration information because the data becomes visible in the URL.

PUT is generally used to update or replace an existing resource.

DELETE is used to remove a resource from the server.

Correct Registration Form

```
<form action="/register" method="POST">
```

```
  <label for="name">Name:</label>
```

```
  <input type="text" id="name" name="name" required>
```

```
  <br><br>
```

```
  <label for="email">Email:</label>
```

```
  <input type="email" id="email" name="email" required>
```

```
  <br><br>
```

```
  <label for="password">Password:</label>
```

```
  <input type="password" id="password" name="password" required>
```

```
  <br><br>
```

```
  <button type="submit">Register</button>
```

```
</form>
```

For secure transmission, the website must use HTTPS.

3. Proper Semantic Element for Navigation

Given Options

A.

`<div>`

B.

`<nav>`

C.

`<section>`

D.

`<header>`

Correct Answer

Option B:

`<nav>`

Justification

The `<nav>` element is the correct semantic HTML5 element used to define a navigation section.

It is normally used for:

- Main website menu
- Page links
- Category links
- Previous and next navigation
- Sidebar navigation

Using `<nav>` helps browsers, search engines, and screen readers understand that the content contains navigation links.

A `<div>` is a general container and does not describe the purpose of its content.

A `<section>` groups related content but is not specifically meant for navigation.

A `<header>` contains introductory content such as a heading, logo, or navigation area, but the navigation links themselves should be placed inside a `<nav>` element.

Correct Navigation Structure

`<nav>`

`<ul>`

`<li><a href="index.html">Home</a></li>`

`<li><a href="about.html">About</a></li>`

`<li><a href="services.html">Services</a></li>`

```
<li><a href="contact.html">Contact</a></li>
</ul>
</nav>
```

Benefits of Using <nav>

- Improves HTML structure.
 - Supports accessibility.
 - Helps screen-reader users.
 - Improves search engine understanding.
 - Makes the code easy to read and maintain.
-

4. Correct XML Syntax

Given Options

Option A

```
<note><to>User</to><from>Admin</from>
```

Option B

```
<note><to>User</to><from>Admin</from></note>
```

Option C

```
<note><to>User</from></note>
```

Option D

```
<note to="User"></note>
```

Correct Answer

Option B:

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Explanation

Option B is correct because:

- It contains one root element called <note>.
- Every opening tag has a matching closing tag.
- The elements are properly nested.
- The tags are complete.

- The syntax follows XML rules.

Errors in the Other Options

Error in Option A

```
<note><to>User</to><from>Admin</from>
```

The root element `<note>` is not closed.

The corrected version is:

```
<note><to>User</to><from>Admin</from></note>
```

Error in Option C

```
<note><to>User</from></note>
```

The `<to>` element is opened, but it is incorrectly closed using `</from>`.

Opening and closing tag names must match exactly.

The corrected version is:

```
<note><to>User</to></note>
```

Error in Option D

```
<note to="User"></note>
```

The closing `</note>` tag is incomplete because the final angle bracket `>` is missing.

The corrected version is:

```
<note to="User"></note>
```

Important XML Rules

XML documents must follow these rules:

1. Every XML document must have one root element.
2. Every opening tag must have a closing tag.
3. Tags must be properly nested.
4. XML tags are case-sensitive.
5. Attribute values must be enclosed in quotation marks.
6. Tag names must match exactly.
7. Special characters must be escaped when required.

5. Correct Protocol for Web Communication

Given Options

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Correct Answer

Option B: HTTP

Justification

HTTP stands for **Hypertext Transfer Protocol**.

It is the standard protocol used for communication between:

- Web browsers
- Web servers
- Web applications
- Web APIs

When a user opens a website, the browser sends an HTTP request to the server. The server processes the request and sends an HTTP response back to the browser.

Example HTTP Request

GET /index.html HTTP/1.1

Host: www.example.com

Example HTTP Response

HTTP/1.1 200 OK

Content-Type: text/html

Content-Length: 1500

For secure communication, HTTPS should be used.

HTTPS stands for **Hypertext Transfer Protocol Secure**. It encrypts the communication between the browser and the server.

Purpose of Other Protocols

FTP

FTP stands for File Transfer Protocol.

It is mainly used to transfer files between computers.

SMTP

SMTP stands for Simple Mail Transfer Protocol.

It is used to send emails.

SNMP

SNMP stands for Simple Network Management Protocol.

It is used to monitor and manage network devices.

Therefore, HTTP or HTTPS is the correct protocol for web communication.

Complete HTML and CSS Implementation

The following code combines video embedding, semantic navigation, a registration form, and CSS styling.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta
```

```
    name="viewport"
```

```
    content="width=device-width, initial-scale=1.0"
```

```
>
```

```
<title>Startup Web Application</title>
```

```
<style>
```

```
  body {
```

```
    font-family: Arial, sans-serif;
```

```
    margin: 0;
```

```
    background-color: #f4f4f4;
```

```
  }
```

```
  header {
```

```
    background-color: #333333;
```

```
    color: white;
```

```
    padding: 20px;
```

```
    text-align: center;
```



```
}
```

```
nav {  
    background-color: #555555;  
    padding: 10px;  
}
```

```
nav ul {  
    list-style-type: none;  
    margin: 0;  
    padding: 0;  
    text-align: center;  
}
```

```
nav li {  
    display: inline;  
    margin: 15px;  
}
```

```
nav a {  
    color: white;  
    text-decoration: none;  
}
```

```
main {  
    width: 80%;  
    margin: 20px auto;  
}
```

```
section {  
    background-color: white;  
    padding: 20px;
```

```
margin-bottom: 20px;  
border-radius: 8px;  
}
```

```
video {  
    max-width: 100%;  
}
```

```
label {  
    display: block;  
    margin-top: 10px;  
}
```

```
input {  
    width: 100%;  
    padding: 8px;  
    margin-top: 5px;  
    box-sizing: border-box;  
}
```

```
button {  
    margin-top: 15px;  
    padding: 10px 20px;  
    background-color: #333333;  
    color: white;  
    border: none;  
    cursor: pointer;  
}
```

```
button:hover {  
    background-color: #555555;  
}
```

```
    footer {
        background-color: #333333;
        color: white;
        text-align: center;
        padding: 10px;
    }
</style>
</head>

<body>

<header>
    <h1>Startup Web Application</h1>
</header>

<nav aria-label="Main navigation">
    <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#video">Video</a></li>
        <li><a href="#register">Register</a></li>
    </ul>
</nav>

<main>

<section id="home">
    <h2>Welcome</h2>

    <p>
        Welcome to our modern web application developed using
        standard HTML5 and CSS.
```

</p>

</section>

<section id="video">

<h2>Promotional Video</h2>

<video width="640" height="360" controls>

<source src="video.mp4" type="video/mp4">

<source src="video.webm" type="video/webm">

Your browser does not support the video element.

</video>

</section>

<section id="register">

<h2>User Registration</h2>

<form action="/register" method="POST">

<label for="name">Full Name:</label>

<input

type="text"

id="name"

name="name"

required

>

<label for="email">Email Address:</label>

<input

type="email"

```
        id="email"
        name="email"
        required
    >

    <label for="password">Password:</label>

    <input
        type="password"
        id="password"
        name="password"
        required
    >

    <button type="submit">Register</button>

</form>
</section>

</main>

<footer>
    <p>&copy; 2026 Startup Web Application</p>
</footer>

</body>
</html>
```

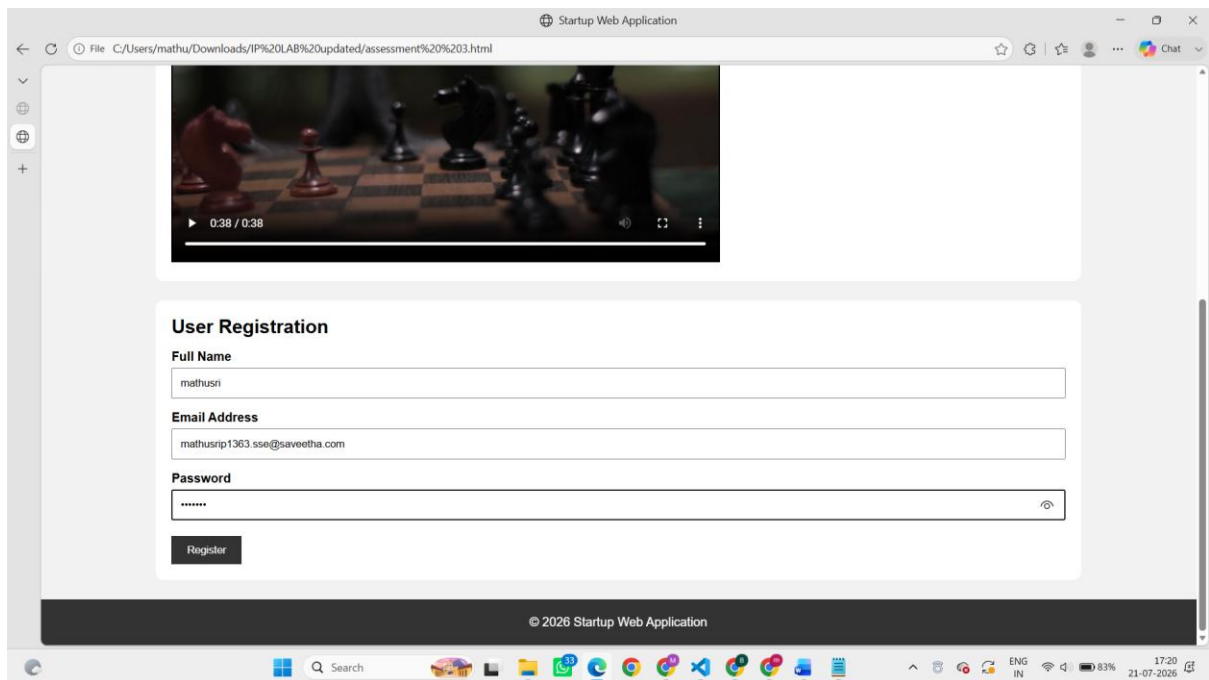
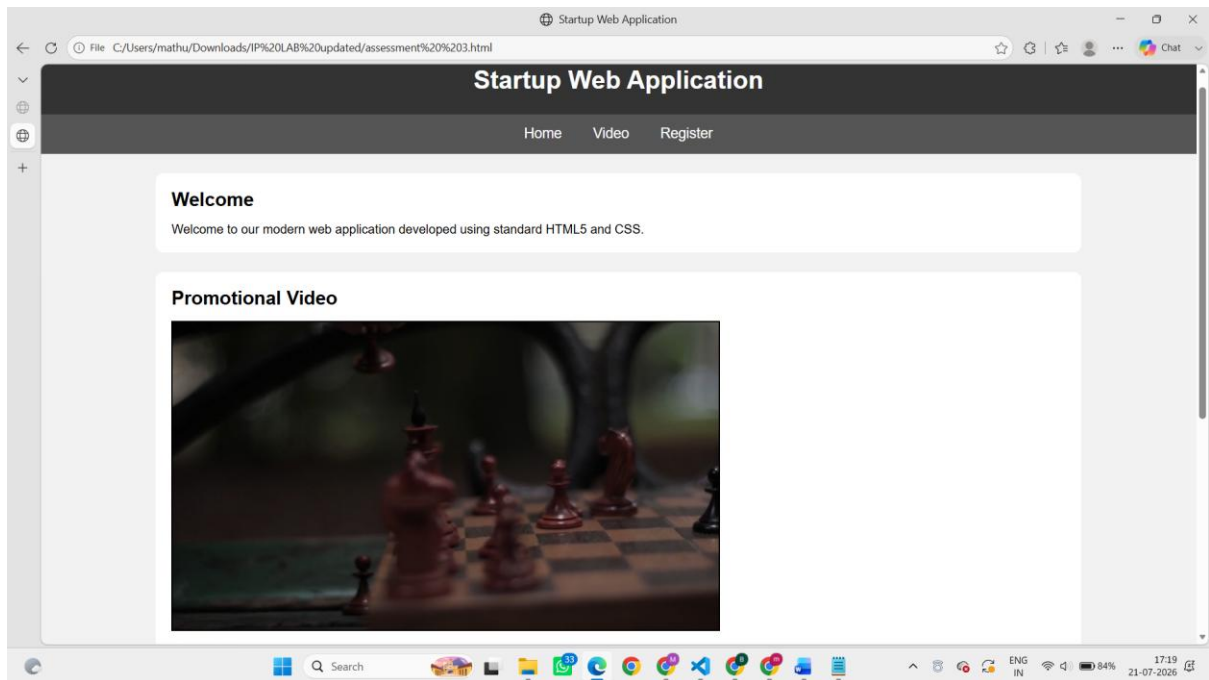
Expected Output

The web page displays:

- A header containing the application title.
- A semantic navigation menu.

- A welcome section.
- A promotional video with playback controls.
- A user registration form.
- A styled registration button.
- A footer containing copyright information.

The page uses CSS to provide a clean, simple, and user-friendly design.



Final Answers Summary

Question	Correct Option	Answer
Video embedding	B	<video>
Form submission	B	POST
Navigation element	B	<nav>
XML syntax	B	Properly closed <note> structure
Web communication	B	HTTP

Conclusion

The case study demonstrates the correct application of important Internet Programming concepts.

The HTML5 <video> element is used to embed promotional videos. The POST method is suitable for registration forms because it sends data in the request body. The <nav> element provides semantic navigation. XML documents must have correctly matched and nested tags. HTTP and HTTPS are used for communication between web browsers and servers.

Following these standards improves security, accessibility, browser compatibility, code readability, and overall web application quality.

Code of Conduct

I, Mathusri-192311363 (**Name / Register Number**), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mathusri P